Exploring Procedural Understanding and AI Agents:

Using a USFCA inspired Chatbox to Support Procedural Understanding in Academic Planning

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CS 486, CS 686

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1 Introduction

Planning an academic journey is a critical yet time-consuming task for university students. It requires careful consideration of prerequisites, scheduling, and course availability across semesters. This process can become complex for students at the University of San Francisco (USFCA) as they navigate the extensive course catalog and align their academic plans with personal goals. Students need to know which exact modules they need to take. Many turn to academic counselors or course directors for guidance as pre-requisite chaining and adding personal goals can be quite challenging to make a concrete academic plan so that they can graduate within 4 years. To address these challenges, this study explores how artificial intelligence can streamline academic planning and enhance the student experience.

This research focuses on the development and evaluation of an enhanced AI chatbot (similar to existing Large Language model chatbots such as ChatGPT) specifically designed to assist USFCA students in creating efficient and personalized academic plans for Computer Science undergraduates, by inputting prompts by the student and the AI chatbot generates a response. Unlike generic AI tools, this chatbot integrates a Retrieval-Augmented Generation (RAG) system, which uses the most recent scraped data from the USFCA course catalog, then calculates the most relevant information to create a prompt to send to a large language model so it can use that data instead of the out-dated scraped data it generally uses for its generic responses. The chatbot aims to provide accurate responses that uses up-to-date scraped USFCA course catalog data for students to help build their Computer Science Undergraduate course plan which would have the correct course chaining and will allow them to graduate within 4 years.

The study compares the enhanced chatbot powered by RAGs with a baseline chatbot that

lacks this advanced functionality, examining their respective impacts on academic planning for undergraduate computer science students. By evaluating the chatbot's effectiveness in improving the planning process, this research aims to contribute to the development of innovative, student-focused tools that enhance academic success and streamline university resources.

The study compares the enhanced chatbot powered by RAG implementation with a baseline chatbot. The differences between the 2 are:

- **Baseline Agent** The Baseline agent uses an API key from OpenAI, once the user enters a prompt, the prompt is sent to OpenAI/ChatGPT and then it sends a response, which is then rendered to the user via using HTML/CSS/JavaScript. Effectively it's a shell/skin for ChatGPT.

- **Enhanced AI Agent** The Enhanced AI agent also uses the API key like the baseline from OpenAI, however before the prompt is sent via API, the following is done:

    1. **Course Catalog Scraping** A previous program was created that scraped the USFCA Computer Science Undergraduate Course Catalog Link: Computer Science Program Overview it also scraped the content of the pages for the courses which had links for each course (as it contained the pre-requisites required for course chaining). For example, the lower division for Computer Science CS 111 page would be scraped as it contained more information on the course Link: CS 111

    2. **Chunking, Embedding and exporting** Once the program gathers all the data, it would then split all of the text into chunks of 500 characters with 100 character

overlap, it would remove new lines, and any other unnecessary spaces to have as much relevant information as possible. The vector embeddings would then be created for each chunk using OpenAI's 'text-embedding-ada-002' model. Then the keys (chunkText and embedding) and values would be stored on a MongoDB database for later retrieval

3. **User Query, retrieval and cosine similarity** During the interaction session the user would enter their prompt, and all the vectors in MongoDB would then have a cosine similarity function checked against the user prompt. A list is created from the most relevant chunks from the user prompt to the least relevant.

4. **Token count calculation** There is a limit to how many characters can be sent to OpenAI API which uses a token count system. It's a complicated calculation, but using their documentation; 1 token is approximately 4 characters, so the prompt would be limited to 4096 characters. So using a function, all the most relevant chunks would be added including the character count of the prompt from the user and the template which is outlined below.

5. **Prompt Template and posting to API** The prompt template is as follows: "Answer the question based on the context below. If you can't answer the question, reply with the data in the prompt and from your scraped data of the USFCA website." + "Context [The chunks most relevant to the user query]" + "Question [user prompt] . The query is then sent to OpenAI's API

6. **Response** OpenAI's API would then create a response based on the information sent to it, and then the text would be formatted to include spaces, bold text, heading text, etc. for it to be easily read by the user.

Additional functionalities

- **Previous chat history** - Each participant will have their assigned ID, and all of the chat inputs that the users have entered will be stored on MongoDB and displayed in the chat interface whenever they go to the AI chatbot page.
- **Message and data collection** - The program collects clicks, hovers, and messages sent to the user as well as the responses from OpenAI.

The below figure shows the user interface of the chatbot:

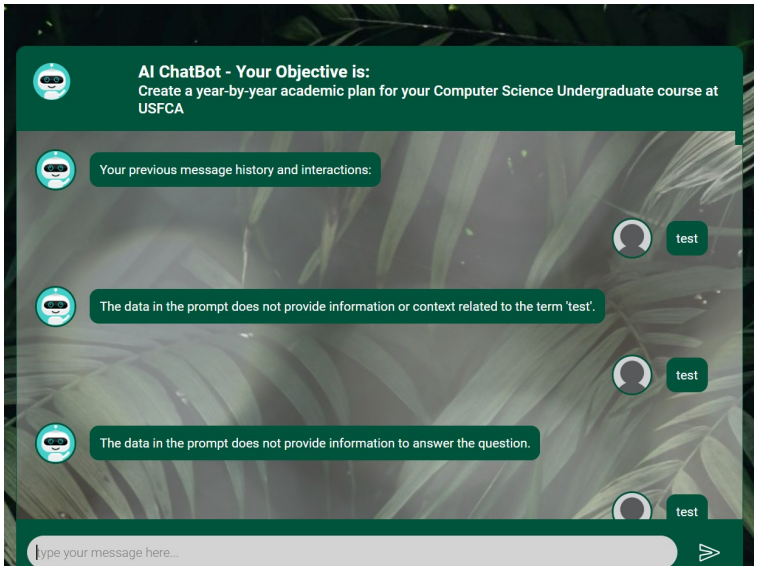


Figure 1: AI Agent interface

Both of the agents had the same exact interface, the only difference is that the back-end of the Enhanced AI agent would have the RAG implementation which added its data to the user prompt.

1.1 Methodology

- **A**n email is sent to each participant containing their user ID and the link to the study: Link: Study Link
- **T**hey enter their ID on the homepage, if valid, it navigates them to the study page that looks like below:

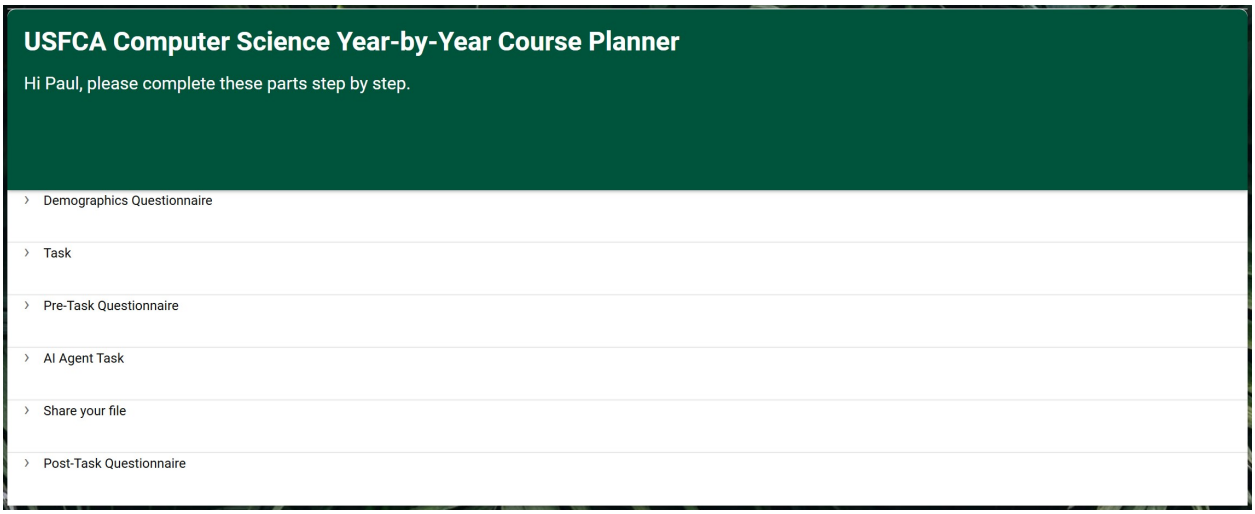


Figure 2: Study process page, each section expands to give them full instruction on each step

- They firstly do a demographics questionnaire
- Next they will create an academic plan on Google Sheets using the USFCA website
- They will be asked to do a pre-task questionnaire
- Afterwards depending on their ID - if the third letter of their ID starts with 'l' then they will be directed to the enhanced AI agent. If it starts with 'f' they will be directed

to the baseline agent. Participants will enter prompts that they think will give them an accurate academic plan for a Computer Science Undergraduate Major, they will create a new tab on the Google Sheet that contains the academic plan they created using the AI agent

- They share their Google Sheet file which then gets stored as a data file on MongoDB for later analysis

- Lastly participants will complete a post-task questionnaire.

1.2 Contributions

- **Paul Gedrimas** - Creation of the back-end, interface, and study workflow, raw data extraction, data cleaning, and running of statistical tests.

- **Lisa Jurca** - Research question and hypothesis, Qualtrics questionnaires (demographics, pre-task, post-task), building individual links for study tasks, drafting report outline, analyzing data for concluding future directions, conclusions, and limitations.

- **Cynthia Carnero** - Helping to address questions for the pre-task and post-task, contributing to writing the reports, analyzing the results, explaining the study design and adding key takeaways.

2 Research Questions

This research focuses on evaluating the effectiveness of the enhanced AI chatbot in addressing key challenges faced by USFCA Computer Science undergraduate majors during academic planning and registration. By integrating Retrieval-Augmented Generation (RAG) and using

an up-to-date USFCA course catalog, this AI chatbot aims to improve the efficiency, confidence, and accuracy of the Academic plans it generates. The following research questions explore the agent's impact on course catalog navigation, career guidance, and personalized course planning. Each research question is accompanied by specific hypotheses, independent and dependent variables, as well as justifications.

2.1 RQ1: Does the AI agent influence the participant's interactions and behaviors?

H1: Students using the Enhanced AI agent (X) will have a different rate of interactions and behaviors (Y) than using the baseline agent.

- **Independent Variable (X):** Use of the AI agent.
- **Dependent Variable (Y):** Participant Interactions and Behaviors, which include: the amount of messages sent to Agents and duration of interaction (calculated from the first prompt to the last response)

Because the enhanced AI will have different prompt outputs, the participants' behaviors may be different from using the baseline AI, which could influence the amount of messages being sent and the duration of the interaction. During testing, prompts from the Enhanced AI are much more condensed, more specific, and less text that attempts to make the AI seem more 'human'. The hypothesis is that because of this, message quantity and session duration would be affected.

2.2 RQ2: Does the AI agent provide useful and relevant career advice to students?

H2: Students using the Enhanced AI agent (X) will report a different rate of confidence (Y) in the relevance and usefulness of career advice than using the Baseline AI agent.

- **Independent Variable (X):** Use of the AI agent.
- **Dependent Variable (Y):** Confidence in the relevance/usefulness of advice (measured via a Likert scale).

Because of the up-to-date data (scraped from the USFCA website on 11/20/2024) that will be added to the prompt for the enhanced AI, users will get different responses from the AI, which in turn will affect their perception and evaluation of the tool.

The following parameters were measured:

- **User confidence in the effectiveness of creating an Academic Plan by AI chatbot** the question in the Post-Task questionnaire below was asked using a Likert Scale 1-5 (1 - Strongly Disagree, 5 - Strongly Agree): *Using the chatbot was effective in the task of creating an academic plan.*
- **User confidence in accuracy** the question in the Post-Task questionnaire below was asked using a Likert Scale 1-5 (1 - Strongly Disagree, 5 - Strongly Agree): *The chatbot's responses were accurate in creating an academic plan.*
- **User confidence in accuracy** The question in the Post-Task questionnaire below was asked using a Likert scale of 1-5 (1 - Strongly Disagree, 5 - Strongly Agree): *The chatbot's responses were accurate in creating an academic plan.*

- **Response relevancy/response in accurate context based on user prompt** The question in the Post-Task questionnaire below was asked using a Likert Scale 1-5 (1 - Strongly Disagree, 5 - Strongly Agree): *The chatbot's responses were relevant.*

- **Ease of use and ease of navigation** The question in the Post-Task questionnaire below was asked using a Likert Scale 1-5 (1 - Strongly Disagree, 5 - Strongly Agree): *The chatbot was easy to navigate and interact with.*

Because these questions are user evaluations, they don't necessarily measure the true accuracy of the academic plans they created i.e. they may feel confident that their academic plan is correct but can then realize that in fact, it has errors. However, RQ2 is an important metric because users need to feel that the information is accurate and easy to use, and therefore enjoyable, which is a crucial metric if such a tool would ever be deployed publicly.

RQ3: Does the AI agent help CS majors create a course map/outline tailored to their academic goals?

H3: The AI agent (X) will generate an accurate course map/outline (Y) for CS majors tailored to their academic needs.

- **Independent Variable (X):** Use of the AI agent.

- **Dependent Variable (Y):** Accuracy of the course map/outline (measured through evaluation by academic counselors).

This metric was measured by asking a Computer Science Course Professor to develop an accurate academic plan for a CS Undergraduate student.

Enter the classes you took or plan to take each semester:

Fall '24	Spring '25	Fall '25	Spring '26
Introduction to Computer Science I - CS 110 - 01	CS 111 - prq CS 110	Math 202 - prq MATH 201	CS 245 - prq MATH 201 and CS 110
Math 109 Calculus	MATH 201 - prq CS 110	CS 112 - prq CS 110	CS 221 - prw CS 112

Fall '26	Spring '27	Fall '27	Spring '28
CS 272 - prq CS 245	CS 490 - prq CS 272		
CS 315 - prq CS 221	CS 326/336 - prq CS 221		
CS elective one	CS elective 2		

Figure 3: Academic plan created by Computer Science Course Professor

The orange highlighted courses are the necessary courses a student needs to do that will help them graduate. These are the most crucial courses a CS student needs to take.

Because each semester has 2 courses that need to be taken and there's 6 semesters, a total of 12 points will be used as a top score.

The simple calculation is whenever the participant has the correct course in the semester that matches the one in Figure 3, they would get awarded +1 point, with a maximum of 12 points.

The Google Sheets that the participants created using the responses from the AI agent will be assessed and scored.

Because the baseline and enhanced AI agents use different data to create their responses;

it's hypothesized that there will be a difference in the score and, therefore accuracy of the academic plans of participants who used the baseline and enhanced AI agents.

2.3 Participants

Undergraduate and Graduate students enrolled in the course CS 486 and CS 686 will be the participants of this study. These students are the target demographic for this study. This is because they are actively engaged in academic planning and registration plans. They will; in the future have to enroll in a class(es), or have done so already. Their experiences will provide valuable insights into each chatbot's effectiveness in academic course structuring. This includes navigating the course catalog, receiving career advice, and creating personalized course outlines.

2.4 Data Collection

A variety of data points will be collected to evaluate the enhanced AI agent's performance. The following data will be logged during interactions with the AI Chatbot:

- **Task Completion Time:** The total time from the first user prompt to the last response generated by each agent.
- **Accuracy of Answers:** The correctness of the AI's responses, by using and scoring their academic plans.
- **Usefulness of Ideas Generated:** The relevance of the chatbot's suggestions for career advice or academic planning, assessed through post-task surveys.
- **System Logs:** Conversation history logged on MongoDB, including all the prompts, their date-time stamps, responses, and overall conversation history.

This data will then be enumerated and quantified using RStudio for analysis and interpretation.

2.5 Questionnaires & Assessments

The participants will complete three questionnaires: a Demographics Questionnaire, a Pre-task Questionnaire, and a Post-task Questionnaire.

In the Demographics Questionnaire, participants will be asked to provide information such as their age, gender, year in university, and academic major. This will help gather specific insights into the participants' background, particularly for computer science students participants. Although demographic analysis was outside the scope of the research questions this paper studies, it can be used for various other analyses for future research.

Link: Demographics Survey

The next questionnaire that participants will complete is the Pre-task Survey. This survey will assess their familiarity with academic planning processes as well as their experience and comfort level with artificial intelligence tools. By gathering this information, we will gain insights into participants' knowledge of both academic planning and AI. Understanding their familiarity with these areas will allow us to tailor our analysis and interpret how prior experience might influence their interactions with the chatbot.

Link: Pre-task Survey

The final questionnaire the participants will complete is the Post-task Survey. This is designed to evaluate the metrics outlined above. First, it will measure the efficiency of the chatbot as perceived by the user in tasks such as course selection or academic planning. Next, the participants will be asked to evaluate the accuracy of the information provided by the enhanced chatbot. This will determine if the information was reliable and aligned with

their expectations. The survey will then explore whether the chatbot saved the user time compared to traditional methods of academic planning, which typically involve browsing multiple web pages and course catalogs.

The questions from the Post-Task that will test RQ2's hypothesis are the following:

- *Using the chatbot was effective in the task of creating an academic plan.*
- *The chatbot's responses were accurate in creating an academic plan.*
- *The chatbot's responses were relevant.*
- *The chatbot was easy to navigate and interact with.*

Lastly, the survey will gauge the participants' future usability by asking if they would choose to use the chatbot again. Together, these questions will provide an insight into the chatbot's performance and its impact on the user experience.

Link: Post-task Survey

2.6 Tasks

The participants will create an academic 4-year plan for the Undergraduate Computer Science Course. The course needs to follow the correct prerequisites and, therefore be chained correctly.

Scenario: Using the currently available data on the USFCA website and using the AI agent, create a year-by-year academic plan for all the semesters and all modules that need to be taken to complete the Computer Science Undergraduate Degree.

Task given to participants: Create a year-by-year academic plan for your Computer Science Undergraduate course at USFCA, This includes all the required modules you will

need to complete in order to graduate and all the modules you will add need to have the proper prerequisites completed in order for it to have the correct chaining and be valid. You will first use the USFCA website to create an Academic Plan, and next, you will use an AI agent to create one.

You will use Google Sheets to create both academic plans.

You are free to create the sheets however you want in terms of style and layout.

2.7 Study Protocol

Participants will follow these steps during the study:

1. **Demographics Survey:** Participants will complete a brief demographics survey, estimated to take 2–3 minutes.

2. **Manual creation of course plan:** Participants will create an academic plan using the USFCA website as their source of data. This task is estimated to take 20-30 minutes. They will split the screen so that one side will be the USFCA website, the other is the Google Sheet where they will create the academic plan from the information they acquire.

3. **Pre-Task Questionnaire:** Participants will complete a pre-task questionnaire asking how difficult it was to create the academic plan manually as well as their experience in AI tool usage.

4. **Chatbot Interaction Task:** With a split screen between Google Sheets and the AI agent, the participants will enter prompts that they believe will help them achieve their objective. They will then enter the data into a separate tab on Google Sheets called 'Agent' to create their academic plan.

(a) **Baseline chatbot Group:** Participants will use a raw API input/output from Chat-GPT 'gpt-4'.

(b) **Enhanced Chatbot Group:** Participants will use the enhanced AI chatbot with RAG implementation of up-to-date course data scraped from the USFCA Computer Science Undergrad Program Overview/Catalog.

Once complete, they will submit their Google Sheet file using a form on the main study page. This task is estimated to take 10-20 minutes.

5. **Post-Study Survey:** After completing the task, participants will fill out a post-study survey to share their experience using the chatbot to create academic plans. This task is estimated to take 3–5 minutes.

The link to the full study workflow can be accessed with the link below, please enter a participant ID for access - 63ld87d7g

Link: Study Workflow

2.8 Study Design

The study will employ a between-subjects design, dividing a sample of approximately 12 participants into two groups. One group will perform a task using the baseline chatbot, while the other group will complete a similar task using the enhanced chatbot. Participants will be randomly assigned to one of the groups, ensuring that the assignment process is unbiased. This randomization ensures that neither the participants nor the investigators can predict or influence the group to which any participant is assigned, maintaining the integrity of the study. A simple random function in Excel was used to assign participants to either the baseline or enhanced agent. Based on their ID; if the 3rd character contained

an 'f', the link would dynamically change to the baseline agent, while if the third character was 'l' the link would change to the enhanced AI agent.

Following this approach offers several benefits:

1. **Simplifies the Design:** Participants would receive an email containing their participant ID, and the same website will dynamically change links for their surveys as well as links to the baseline/enhanced agents all dependent on their ID. This helps reduce prompting by principal study investigators as they should be able to do each section independently.

2. **Allows Direct Comparisons:** By separating the groups, the study can directly compare the performance between the baseline and enhanced chatbots without interference from repeated exposure to similar tasks.

2.9 Data Analysis

This study incorporates both quantitative, but the scope of this study is to measure quantitative data. The qualitative data could be analyzed for other studies or used as inspiration for new studies. This will be discussed more comprehensively in the conclusion.

2.9.1 Quantitative Analysis

For the analysis of the quantitative data, we will use descriptive statistics, such as the mean and standard deviation, to understand how the data is distributed. Additionally, we will examine whether the data follows a normal, bimodal, or multi-modal distribution.

Moreover, we will test our hypotheses to determine which website performs better: the baseline chatbot or the enhanced chatbot. We will also evaluate whether the performance

differences between the two groups are statistically significant. To achieve this, we will use the p-value as a measure of significance. Specifically, a p-value of 0.05 indicates there is only a 5 percent probability of observing these results if the null hypothesis were true.

To compare the averages of the two groups and assess whether the differences are likely due to random chance, we will use a t-test. Since this study employs a between-subjects design, we will use the Welch Two Sample t-test.

3 Results

RQ1: Does the AI agent influence the participant's interactions and behaviors?

The metrics used for determining the AI agent's influence on the participant's interactions were the number of messages and the duration of use (from the first prompt to the last response from the chatbot).

1. **Messages by Condition:**

Table 1: Messages sent: Welch Two Sample t-test Results

Statistic	**Value**
Data	Messages sent by Condition
t-statistic (t)	-0.63786
Degrees of freedom (df)	4.6414
p-value	0.5537
Alternative hypothesis	True difference in means between group Basic and group Enhanced is not equal to 0
95% confidence interval	[-9.396761, 5.730094]
Mean in group Basic	5.166667
Mean in group Enhanced	7

The analysis of messages exchanged by users with the basic and enhanced AI agents

indicates no significant difference between the two conditions. This may be seen in the t-statistic of -0.638 and a p-value of 0.554 (greater than the significance threshold of 0.05) indicates weak evidence of a difference. The adjusted degrees of freedom (4.641) and confidence interval [-9.397, 5.730], which crosses 0 further support the lack of statistical significance in these results.

The mean number of messages sent to the basic agent was 5.17. The mean number of messages sent to the enhanced AI agent was 7.0, which resulted in a mean difference of 1.83 messages. While users interact slightly more with the enhanced agent on average, the result is low and shows no significant statistical difference.

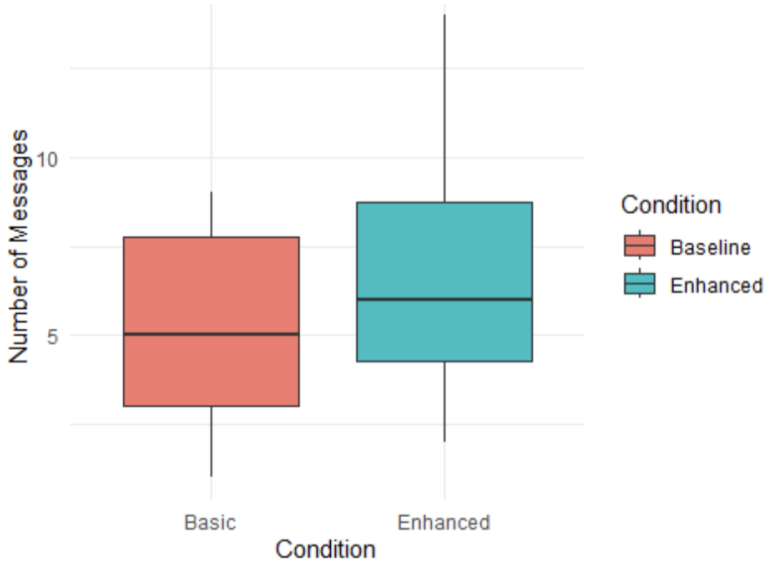


Figure 4: Messages by Condition

2. **Session Duration by Condition:**

Session duration analysis also yielded no significant difference between conditions. The

Table 2: Session Duration: Welch Two Sample t-test Results

Statistic	**Value**
Data	Session Duration by Condition
t-statistic (t)	-0.86572
Degrees of freedom (df)	5.7188
p-value	0.4215
Alternative hypothesis	True difference in means between group Basic and group Enhanced is not equal to 0
95% confidence interval	[-489.4077, 235.8594]
Mean in group Basic	340.4933
Mean in group Enhanced	467.2675

t-statistic of -0.866, a p-value of 0.422, and a confidence interval of [-489.408, 235.859] indicate no strong evidence of a difference.

On average, users spent 340.493 seconds interacting with the basic agent and 467.268 seconds with the enhanced agent. On average participants spent 126.775 more seconds interacting with the enhanced AI agent.

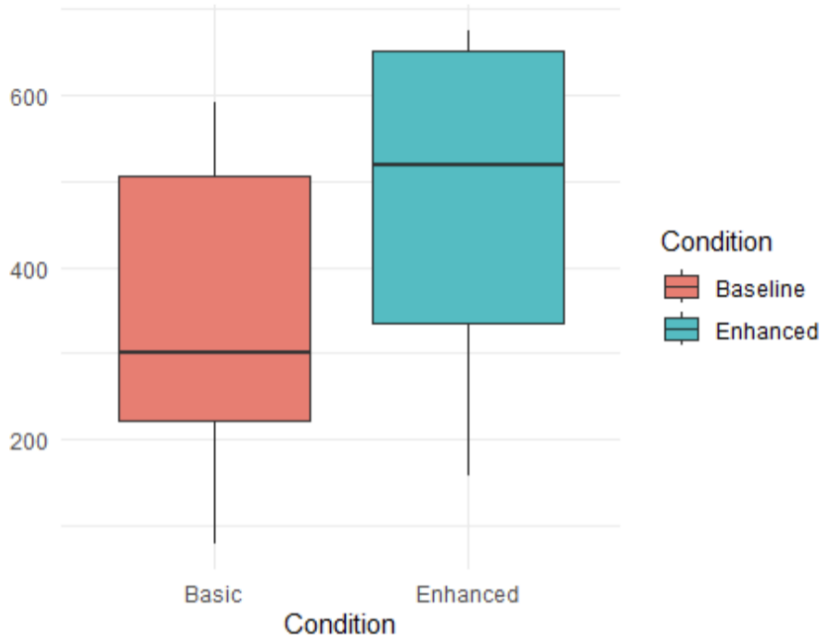


Figure 5: Duration by Condition

However, none of these differences are too significant.

RQ2: Does the AI agent provide useful and relevant career advice to students?

1. **Effectiveness Confidence** Effectiveness Confidence measures how effective users perceived the AI in assisting them with creating an academic plan.

   The Enhanced AI agent had a higher average score (2.25) compared to the baseline AI agent (1.50), suggesting that users felt slightly more confident in its effectiveness. However, this improvement was not statistically significant, as indicated by a p-value of 0.2236.

Table 3: Effectiveness Confidence: Welch Two Sample t-test Results

Statistic	**Value**
Data	Effectiveness Confidence by Condition
t-statistic (t)	-1.4195
Degrees of freedom (df)	0.2236
p-value	1
Alternative hypothesis	True difference in means between group Basic and group Enhanced is not equal to 0
95% confidence interval	[-2.1741142, 0.6741142]
Mean in group Basic	1.50
Mean in group Enhanced	2.25

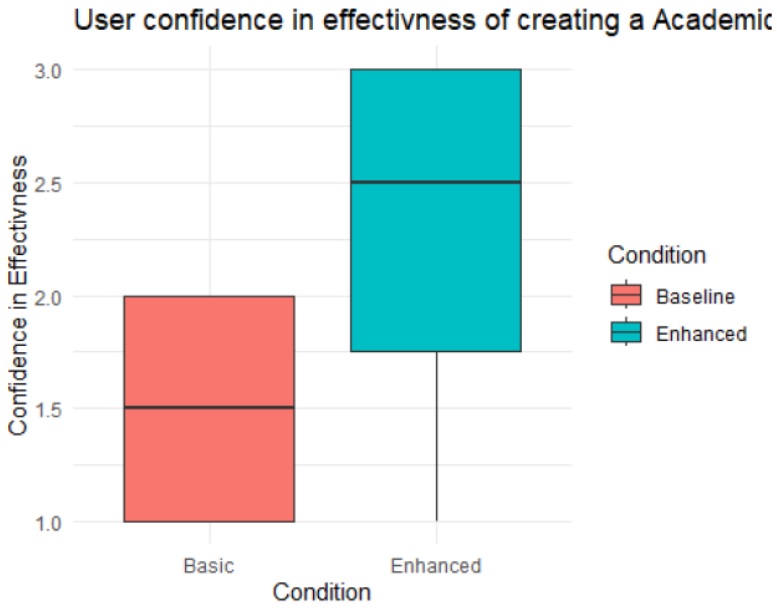


Figure 6: Effectiveness of creating an Academic Plan

Additionally, the boxplot provides insights into user confidence in the effectiveness of creating an academic plan using the baseline and enhanced AI agents, based on a scale where 1 indicates "Strongly Disagree" and 5 indicates "Strongly Agree" based on the question "Using the chatbot was effective in the task of creating an academic plan".

The enhanced AI agent shows a higher median score, suggesting that users were more likely to agree that it was effective, while the baseline AI median remains closer to the "Disagree" category. Despite this higher median, the Enhanced AI also exhibits a wider range of scores, with some users expressing strong agreement (close to 3/ Neither Agree or Disagree) while others rated it closer to disagreement. The baseline AI, in contrast, demonstrates a narrower range of scores, with responses consistently clustered near "Disagree". Based on all the metrics, there isn't much statistical difference, and could be based on randomness.

2. **Accuracy Confidence** Accuracy Confidence reflects users' trust in the AI's ability to provide accurate responses to their queries.

Table 4: Accuracy Confidence: Welch Two Sample t-test Results

Statistic	**Value**
Data	Accuracy Confidence by Condition
t-statistic (t)	0
Degrees of freedom (df)	7.0231
p-value	1
Alternative hypothesis	True difference in means between group Basic and group Enhanced is not equal to 0
95% confidence interval	[-1.294294, 1.294294]
Mean in group Basic	2
Mean in group Enhanced	2

Both the baseline and Enhanced AI agents received the same average score of 2.0, showing no perceived difference in their accuracy.

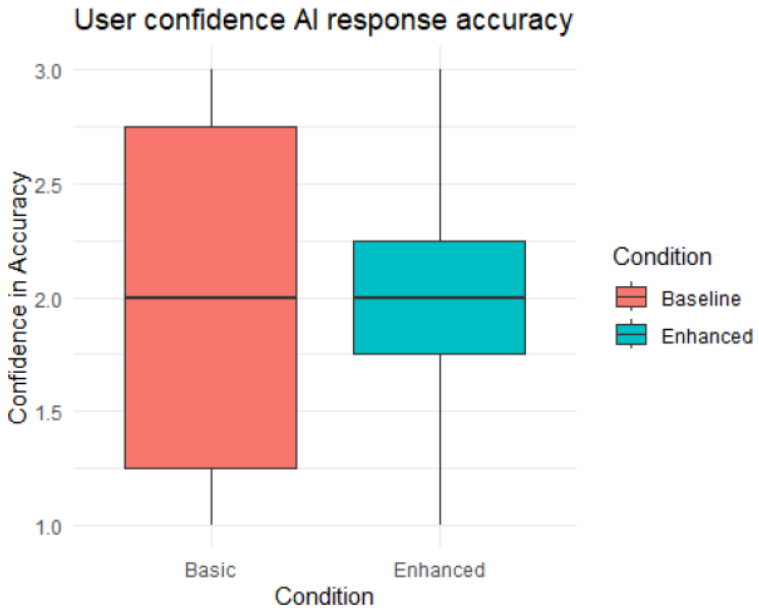


Figure 7: Accuracy by Condition

In addition, the boxplot shows user confidence in the accuracy of responses provided by the baseline and enhanced AI agents, based on a scale where 1 represents "Strongly Disagree" and 5 represents "Strongly Agree" for the question "The chatbot's responses were accurate in creating an academic plan".

Both versions of the AI have a median confidence score of 2, corresponding to "Disagree", indicating that users generally did not find either the baseline or enhanced version accurate in addressing their questions. The Basic AI shows a broader range of ratings, spanning from 1 (Strongly Disagree) to 3 (Neutral), suggesting more variability in user perceptions. In contrast, the enhanced AI has a narrower distribution of scores tightly clustered around 2 (Disagree), reflecting more consistent but still unfavorable opinions about its accuracy.

3. **Response Relevancy** Response Relevancy evaluates whether the AI's responses were relevant to the users' needs.

Table 5: Response Relevancy: Welch Two Sample t-test Results

Statistic	**Value**
Data	Response Relevancy by Condition
t-statistic (t)	1
Degrees of freedom (df)	5
p-value	0.3632
Alternative hypothesis	True difference in means between group Basic and group Enhanced is not equal to 0
95% confidence interval	[-0.5235273, 1.1901939]
Mean in group Basic	2.833333
Mean in group Enhanced	2.500000

Interestingly, the baseline AI agent performed slightly better in this metric, with an average score of 2.83 compared to 2.5 for the Enhanced AI agent. While this result suggests a small preference for the Basic AI's responses, the difference was not statistically significant ($p = 0.363$).

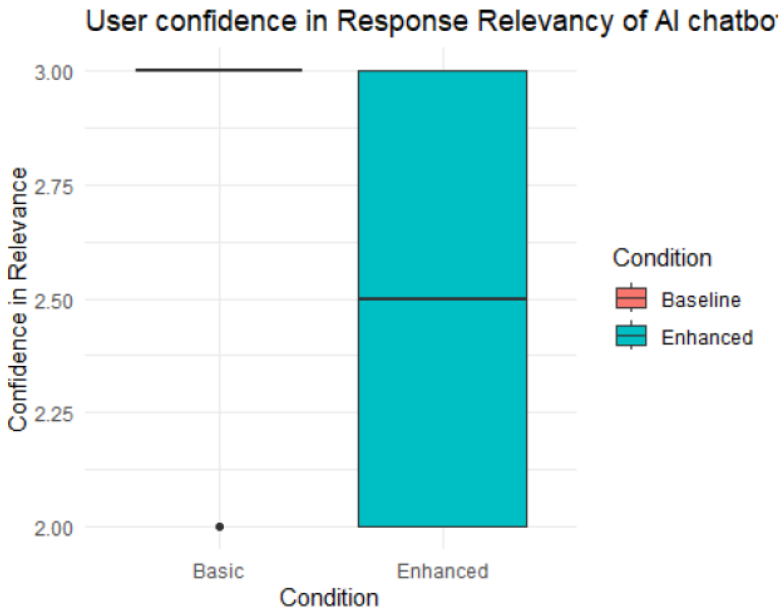


Figure 8: Relevancy by Condition

The boxplot illustrates user confidence in the relevancy of responses provided by the Basic and Enhanced AI agents, measured on a scale where 1 represents "Strongly Disagree" and 5 represents "Strongly Agree" for the question "The chatbot's responses were relevant". The Basic AI agent shows a tightly constrained distribution with a median score of 3 (Neutral), indicating that users generally had a moderate perception of the relevance of its responses. However, the Basic AI includes an outlier at 2 (Disagree), suggesting that a small number of users found its responses less relevant. In contrast, the Enhanced AI exhibits a broader range of scores, with the inter-quartile range spanning from 2.25 (Disagree) to 2.75 (Neutral) and a median at 2.5, slightly lower than the Basic AI's median.

4. **Ease of Use** Ease of Use measures how easy and intuitive participants found the

chatbot to navigate and interact with.

Table 6: Ease of Use: Welch Two Sample t-test Results

Statistic	**Value**
Data	Ease Of Use by Condition
t-statistic (t)	-1.2112
Degrees of freedom (df)	7.1772
p-value	0.2642
Alternative hypothesis	True difference in means between group Basic and group Enhanced is not equal to 0
95% confidence interval	[-3.432954, 1.099621]
Mean in group Basic	1.833333
Mean in group Enhanced	3.000000

The Enhanced AI agent scored higher on average (3.0) compared to the baseline AI agent (1.83), suggesting that users found the Enhanced AI more user-friendly. Despite this apparent improvement, the difference was not statistically significant, with a p-value of 0.264 and a confidence interval of [-3.433, 1.100].

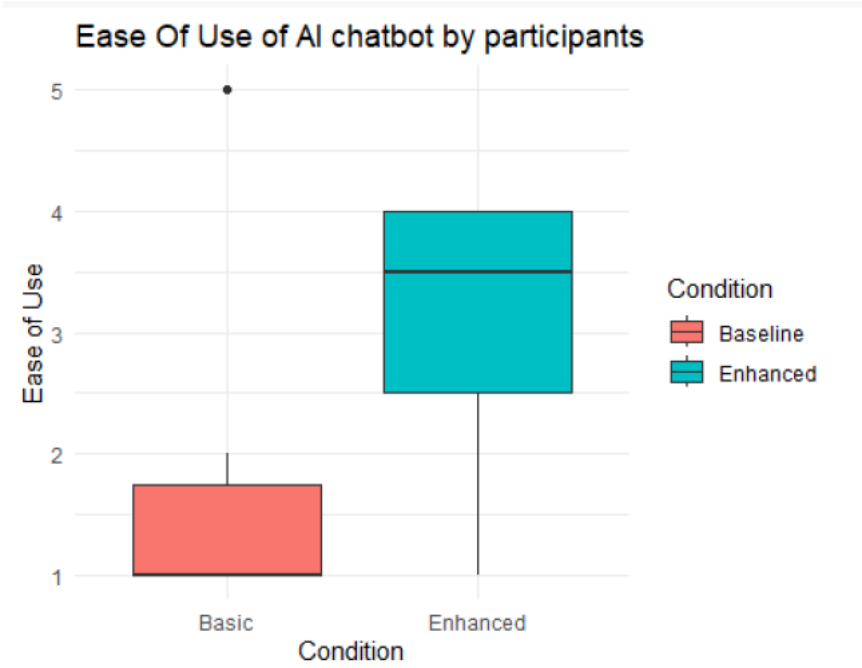
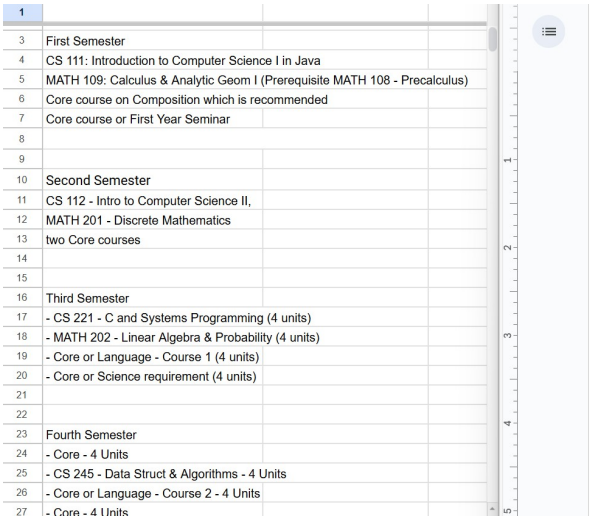


Figure 9: Ease of Use by Condition

The boxplot illustrates the distribution of participants' ratings for the "Ease of Use" of the Basic and Enhanced AI chatbots. The median score for the enhanced AI chatbot is notably higher at approximately 4, compared to the baseline AI chatbot, which has a median score of around 2. The enhanced AI also shows a broader range of scores, with an inter-quartile range (IQR) spanning from 3 to 4, and scores ranging from 1 to 5 for the question "The chatbot was easy to navigate and interact with". In contrast, the baseline AI has a tighter IQR, with most scores clustered between 1 and 2, indicating lower perceived usability. Interestingly, the baseline AI has an outlier at 5, suggesting one participant had an exceptionally positive experience with its usability. No outliers are present for the enhanced AI, indicating more consistent feedback.

3.1 RQ3: Does the AI agent help CS majors create a course map/outline tailored to their academic goals?

The last research question was the analysis of the academic plans created by students using the AI agent were scored against the academic plan created by a Computer Science professor at USFCA.



First Semester
CS 111: Introduction to Computer Science I in Java
MATH 109: Calculus & Analytic Geom I (Prerequisite MATH 108 - Precalculus)
Core course on Composition which is recommended
Core course or First Year Seminar
Second Semester
CS 112 - Intro to Computer Science II,
MATH 201 - Discrete Mathematics
two Core courses
Third Semester
- CS 221 - C and Systems Programming (4 units)
- MATH 202 - Linear Algebra & Probability (4 units)
- Core or Language - Course 1 (4 units)
- Core or Science requirement (4 units)
Fourth Semester
- Core - 4 Units
- CS 245 - Data Struct & Algorithms - 4 Units
- Core or Language - Course 2 - 4 Units
- Core - 4 Units

Enter the classes you took or plan to take each semester:

Fall '24	Spring '25	Fall '25	Spring '26
Introduction to Computer Science I - CS 110 - 01	CS 111 - prq CS 110	Math 202 - prq MATH 201	CS 245 - prq MATH 201 and CS 110
Math 109 Calculus	MATH 201 - prq CS 110	CS 112 - prq CS 110	CS 221 - prw CS 112

Fall '26	Spring '27	Fall '27	Spring '28
CS 272 - prq CS 245	CS 490 - prq CS 272		
CS 315 - prq CS 221	CS 326/336 - prq CS 221		
CS elective one	CS elective 2		

Figure 10: CS Undergraduate Course Schedule created by participant on the left, and the academic plan it's scored against on the right

Every course that was in the correct semester was given +1 point with a maximum of 12 points. The baseline and enhanced AI were then analyzed based on their score.

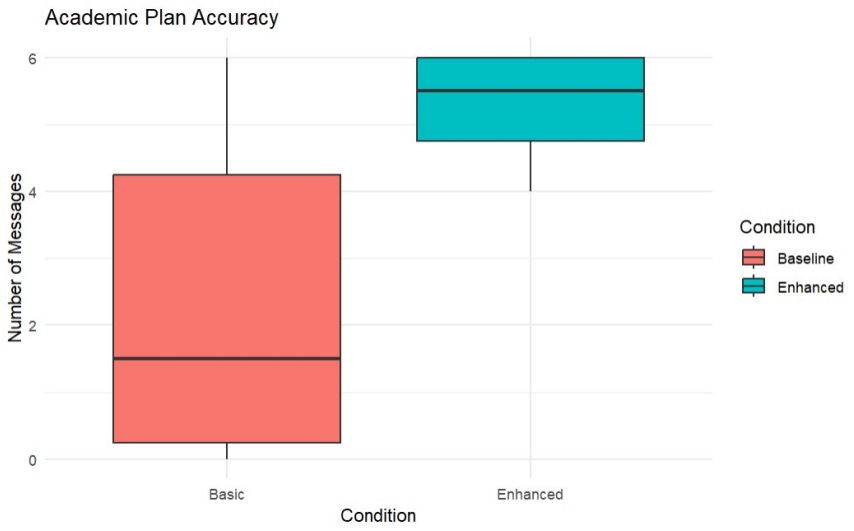


Figure 11: Ease of Use by Condition

Table 7: Welch Two Sample t-test Results

Statistic	**Value**
Data	Plan_Accuracy by Condition
t-statistic (t)	-2.5194
Degrees of freedom (df)	6.7935
p-value	0.04084
Alternative hypothesis	True difference in means between group Basic and group Enhanced is not equal to 0
95% confidence interval	[-5.671156, -0.162177]
Mean in group Basic	2.333333
Mean in group Enhanced	5.250000

The research question is the only one which showed statistical significance in its findings. The p-value of 0.04084 is below the 0.05 standard p-value threshold so the null hypothesis that there is no significant difference can be rejected.

The Confidence interval [-5.671, -0.162] range doesn't include 0, meaning that there is a significant statistical difference between the baseline and AI agent academic plan accuracy.

Research Question	Variable	Baseline	Enhanced	p-value
RQ1	Message Quantity	5.17	7.0	$p = 0.554$
	Session Duration	340.493	467.268	$p = 0.422$
RQ2	Effectiveness Confidence	1.50	2.25	$p = 0.224$
	Accuracy Confidence	2	2	$p = 1$
	Response Relevancy	2.833	2.5	$p = 0.363$
	Ease Of Use	1.833	3.0	$p = 0.264$
RQ3	Plan Accuracy	2.333	5.250	$p < 0.05^*$

Table 8: RQ Summary Results Table (results denoted with * indicate statistical significance)

This shows that the mean of the enhanced group is between 0.16 and 5.67 points higher than the baseline AI agent group.

The enhanced accuracy mean is 5.25 whilst the basic mean is 2.333, so on average the enhanced AI creates an academic plan 2.917 points higher than the baseline AI agent.

Although a larger sample size would be ideal, it does seem that with the up-to-date scraping and RAG implementation of the enhanced AI, it did indeed create more accurate academic plans than the baseline AI agent by a large margin.

4 Discussion

4.1 Interpretation of Findings

The findings from this study provided key insights into the performance and impact of the enhanced AI chatbot across the outlined research questions.

For RQ1, the results showed that the enhanced AI agent had minor increased user activity during interactions, in comparison to the baseline agent. The metrics for message count and duration were higher in comparison on average. This suggests the users were more engaged with the enhanced chatbot, potentially due to the output being different than the generic output of ChatGPT. However, while increased activity is an encouraging sign of usability, it

does not directly translate into greater user satisfaction. Despite the increase in engagement, it was not statistically significant.

The results for RQ2 do not indicate there were significant improvements in user-reported confidence or effectiveness in providing career advice. Participants reported moderate levels of trust in the chatbot's recommendations, but there was not a scalable increase in confidence. This may be reduced to the absence of core requirement data or other vital academic details. Ultimately, this limitation highlights the importance of having a more comprehensive dataset (this will be discussed in the limitations subsection).

The course maps created with the help of the enhanced AI agent outperformed a template course map. When compared, the AI-created plan showed a higher degree of plan accuracy in comparison to the baseline AI agent. However the top score for the enhanced AI agent was 6 out of 12 (just 50%), there is much room for improvement which will be discussed in the limitations subsection).

4.2 Limitations

Although the enhanced AI agent demonstrated notable improvements when creating an academic plan and in course assistance, there are several limitations from development and testing that must be noted:

1. **Scope of RAG implementation:** there were several limitations in the RAG implementation of the enhanced AI:

    - **Missing data:** Due to human error in scraping the USFCA course catalog, CS 110 (Introduction to Computer Science) was missed as a course and instead the enhanced chatbot replaced it with CS 111 for its freshman fall semester course.

Because of this, there would be a cascading error across the whole course schedule. If CS 110 was included in the scraped data, the cascading should have been more accurate, and the enhanced AI theoretically should have performed better.

- **No Core courses:** USFCA requires that all undergraduate students have at least 44 credits in core courses such as language, public speaking, law and order, etc. This data was also not included. If it was, then the academic plans generated would have been more comprehensive and realistic.
- **Vector calculation speed:** The vector-based similarity calculating function based on the user query and the building of the prompt sometimes took about 2 minutes. In turn, there was an increase in wait time for the user when anticipating a response. This could lead to delays in real-time interactions and also detract users.
- **Prompt Limitations:** The AI model utilized OpenAI API keys for querying. This caused the enhanced AI model to have token count restraints, which restricted input length. The limitation skewed the relevance of the responses by truncating contextually important details. This restricted the total quantity of the input string and therefore it may not have included all the data chunks relevant to have all the necessary courses in the prompt.

2. **Participant Priming:** The participants were asked to create a course plan using the USFCA website and course catalog as a pre-task assignment. This may have contributed to skewed results in their final academic plan, as the participants could have used the course website information to assist them in their academic plan made by the AI agent. This was due to the inexperience of the study developers. This could

have been fixed in 2 ways; remove the pre-task assignment completely or just have it as another study variable (baseline AI — enhanced AI — manual creation of plan using USFCA website)

3. **Prerequisites and Placement Tests:** The AI agent did not consistently account for course prerequisites. For example, if the agent is asked for a course's prerequisite very explicitly, the correct answers are given. For example, if the input is "What are the prerequisites for CS315?", the chatbot will respond correctly. However, when the input is more general, such as "What are all of my courses prerequisites?", it will say the query was insufficient. In addition, the agent does not mention placement tests which are needed for several classes or new students coming to the university.

5 Conclusion

5.1 Key Takeaways

The key findings from this study include the following:

1. The enhanced AI chatbot did significantly aid students in creating academic plans tailored to their individual needs. However, due to the model being trained solely on computer science courses, it overlooks certain general education core requirements which account for a large amount of credits to graduate. To further improve this; core courses would be scraped and added as chunks to mongoDB ready to be added as prompts during the user query.

   Even further improvement could be made by adding every course (not just Computer Science) and also including graduate course requirements that would help all students

attending USFCA.

2. Statistical significance is lacking across most metrics. Despite trends favoring the Enhanced AI in Ease of Use and Effectiveness Confidence, no statistically significant differences were observed across most metrics. Again with core classes added, having a faster vector calculation and prompt building should give better more accurate academic plans which in turn build confidence with users and make the experience more pleasant. Improvements in the user interface could also help with this.

3. The limitations of the current OpenAI 'gpt-4' did have a rather small token limit (therefore character limit). To fix this 1 main improvement could be implemented: create a large language model (LLM) from scratch that's hosted on the server and only uses scraped data from the USFCA course catalog. This would allow full control and testing of the LLM to make sure responses do indeed align with user prompts. It would also greatly speed up the responses as the use of APIs in the current design (from vector calculation and prompt sending) slows down the process.

5.2 Research Implications

The development and evaluation of this enhanced AI chatbot for academic planning provided valuable insights into its potential applications, as well as areas of improvement. The following are findings from this study that highlight several implications and practical implementations:

1. **Enhanced Academic Support Tools:** The enhanced AI chatbot demonstrated the potential of truly helping students with accurate academic planning. The improvements mentioned above would greatly help improve the experience.

2. **Bridging Gaps in Accessibility:** Use of an enhanced AI academic chatbot addressing challenges such as navigating complex course catalogs and interpreting career pathways. This might be daunting to new students, as well as confusing for returning students with potentially complex degree evaluations (Bridge programs, Double Majors etc.).

5.3 Future Directions

1. **Have all scraped data from the course catalog for every degree that USFCA provides:** This will allow any student in USFCA to use the chatbot to help them create an academic plan.

2. **Add visualizations and downloadability** A simple addition would allow users to download their academic plans in CSV or Excel format. Visuals showing course prerequisites and minimum grades allowed could be easily implemented with javascript and other web tools. This would allow easy interpretation of a large quantity of data.

3. **Improve speed and UI** Having a built-in LLM and built-in vector creation function and reducing the need for API calls as well as improving the UI (adding loading icons) would greatly improve the speed as well as showing the user more explicitly when the ai agent is generating the prompt.

By providing USFCA students with tailored, efficient, and reliable AI tools, they could navigate their educational journeys with more ease.